Report R9 — Sectors and basins: the weak-component decomposition over the rule space

Tier 1e, open boundaries

QCA Sector Analysis project (Tier 1)

2026-07-29 (`tier1e_version 1e.1`, `obc0`)

Abstract

The dissipative pipeline has so far reported terminal strongly connected components — "recurrent classes" — and their basins. Those are the right object for the *measured* circuit and the wrong object for a sector axis: they do not partition the basis, they are monitored-relative, and the Tier-2 wall grammar predicts sectors rather than attractors. This report adds the weakly connected component (WCC) as a first-class observable, which fixes all three at once: span(WCC) is an enclosure of the channel, invariant under every Kraus operator and therefore exact for the monitored and the unmonitored dynamics alike; WCCs partition the basis; and they are precisely the minimal projectors of the *diagonal* part of the commutant. We sweep all 256 rules at open boundaries, verify the sum rule and the cross-tier consistency assertions, and test the exclusion curve $ab \geq 2$ that a partition forces on the sector map. Two negative results are recorded prominently: the Tier-2 prediction $a_{\mathrm{wcc}} \geq \varphi$ for rule 28 is *false* — the sector count obeys an exact recurrence with base the plastic number $\rho = 1.32472$ — and the basin sum rule turned out to be unanswerable from the Tier-1a archive for 76 units, which we fixed by recomputing them independently. This report covers `obc0` only; `pbc` follows separately.

Contents

1 Three tiers, and why the WCC is the sector-level object

1.1 Vocabulary

The single most important thing this report changes is naming. Throughout the project from here on:

sector	weakly connected component; exact for *both* experiments
monitored attractor	terminal SCC; a property of the measured circuit
channel attractor	block of the fixed-point algebra (Tier 1b / 1d)

A terminal SCC is never a "sector"; a WCC is never an "attractor". Every table and figure caption below states which experiment its numbers describe.

1.2 Why terminal SCCs cannot carry a sector axis

(P1) They do not partition. Transient states belong to no recurrent class, so $\sum_k |A_k| \ll 2^N$ and no sum rule constrains the multiset. The unitary sectors *do* partition, so the existing figures put two different kinds of object on one axis. Rule 22 at $N = 12$ is the extreme case: three monitored attractors, each a single state, covering 3 of 4096 states, against two sectors covering all 4096.

(P2) They are monitored-relative. The transient/recurrent split is a property of the measured chain; the unmonitored channel can keep immortal weight on transient states (R5 rule 22, R7).

(P3) They are not what the analytics predicts. The Tier-2 wall grammar and transfer matrices give sector counts.

1.3 What a WCC is

Definition 1. The one-cycle support graph has nodes $x \in \{0,1\}^N$ and an edge $x \to y$ whenever $\langle y|\Phi_C(|x\rangle\langle x|)|y\rangle \neq 0$. A *sector* is a connected component of this graph read as undirected.

Proposition 1 (enclosure)**.** span(*sector*) *is invariant under every Kraus operator of* Φ_C*, hence invariant for the monitored and the unmonitored dynamics alike.*

The commutant reading is the one that connects this tier to R8. A diagonal operator D commutes with the dynamics iff $D_x = D_y$ whenever the support graph has an edge $x \to y$, i.e. iff D is constant on components. So

$$\dim\{\, D \text{ diagonal} : [D, \Phi_C] = 0 \,\} \;=\; \#\text{components} \;=\; n_{\text{wcc}}, \tag{1}$$

and the graph analysis computes *exactly* the diagonal part of the commutant — no more. That is why it cannot see reflection, or any other symmetry whose eigenvectors are superpositions rather than basis states (R8 §10.3: at $N = 11$ the diagonal part is 7-dimensional inside a commutant of dimension 23 360).

1.4 The algorithm, and one deliberate deviation

A single streaming pass over $\text{succ}(x)$ for $x = 0 \ldots 2^N - 1$, union-find with union by size and path compression, every edge symmetrised by construction ($\text{union}(x, y)$ for each $y \in \text{succ}(x)$). No adjacency storage, no DFS stack. This is the code path the unitary sweep already used, now factored into `graph/wcc.py::weak_components` and shared by both callers.

Deviation, and why. The Tarjan sweep aborts a unit as soon as one component exceeds $f_{\text{erg}} 2^N$. The WCC pass does *not*: aborting leaves an incomplete partition, which would break the sum rule A1 and destroy the exact anchors the exclusion-curve test needs — rule 51 has to come out as one sector of size exactly 2^N, not as "something exceeded half the space". Union-find is cheap enough to always finish, so `ergodic` here is a *classification* derived from $D_{\max}$, and the exponential saving lives in the sweep instead. A second guard: a rule is not written off as ergodic on the strength of a tiny chain, since at $N = 6$ almost anything connects past half of 64 states; the verdict must hold at two consecutive N and never below $N = 10$.

2 The sweep

All 256 rules at `obc0`, swept with N in the *outer* loop so that a run stopped against a wall-clock budget still leaves uniform coverage — every rule with the same N range and therefore comparable fits. Records live in their own store `results_tier1e/`, keyed on $(\text{rule}, N, \texttt{bc})$ and versioned `1e.1`.

A note on versioning. The task asks for the shared `ENGINE_VERSION` to be bumped to `1e.1`. Doing that would make `has_unit()` false for all ∼6000 archived Tier-1a records, so the next sweep would recompute the lot — which the same section explicitly forbids. The version bump therefore lives on the new tier, exactly as Tier 1d does. Nothing in Tier 1a/1b/1d is recomputed or invalidated.

N	rules covered	2^N
6	256	64
7	256	128
8	256	256
9	256	512
10	256	1 024
11	256	2 048
12	256	4 096
13	256	8 192
14	256	16 384
15	256	32 768
16	256	65 536
17	73	131 072
18	3	262 144

Coverage is uniform: all 256 rules at every $N = 6 \dots 16$, so the descriptors of §5 are all fitted on the same domain. A targeted set of 30 diagnostic rules was carried to $N = 17$–18 afterwards; the headline map deliberately does *not* use those extra points, since a rule fitted on a longer window is not comparable with one fitted on the uniform window.

3 Ground truths

All of the following are regression-tested in `tests/test_wcc.py`.

- **Rule 204** (`IIII`): 2^N sectors of size 1; $a = 2$, $b = 1$.
- **Rule 51** (`VVVV`): one sector of size 2^N; $a = 1$, $b = 2$.
- **Rule 150** `obc0`: sectors are the domain-wall-number eigenspaces, $|S_w| = \binom{N+1}{w}$ on even w. The computed values reproduce the reference series for $N = 8 \dots 17$ exactly — #sectors $5, 6, 6, 7, 7, 8, 8, 9, 9, 10$ and $D_{\max}$ 126, 210, 462, 924, 1716, 3003, 6435, 12870, 24310, 43758.
- **Rule 156** `pbc`: $n_{\text{wcc}} = \text{Lucas}(N) + 1$, verified $N = 6 \dots 12$.
- **Rule 22** `pbc`, even N: three monitored attractors, all of size 1, against $n_{\text{wcc}} = 2$; the transient fraction runs $0.9531, 0.9883, 0.9971, 0.9993$ at $N = 6, 8, 10, 12$ and $\to 1$.

A naming trap worth recording. Wolfram numbers are not classical ECA rules in this encoding. W_{90} has symbol tuple `DEED` and is *dissipative*; assertion A2 must not be applied to it. Its $N = 8$ `obc0` numbers make the P1 point cleanly: sector sizes $112, 112, 16, 16$ summing to 256, against terminal-SCC sizes $7, 7, 1, 1$ summing to 16.

4 Rule 28: the wall-transparency prediction fails

The task predicts $a_{\text{wcc}} \geq \varphi$ for rule 28 (`IIVD`), on the grounds that it inherits the 01-wall grammar of rule 156, and asks for a prominent flag if that is violated. **It is violated.** The `obc0` sector count is

$$n_{\text{wcc}}(N) = 11, 15, 20, 27, 36, 48, 64, 85, 113, 150, 199 \qquad (N = 6 \dots 16),$$

which satisfies the exact integer recurrence

$$a_n = a_{n-1} + a_{n-2} - a_{n-4}, \tag{2}$$

verified out of sample on every available term. Its characteristic polynomial factors as $x^4 - x^3 - x^2 + 1 = (x-1)(x^3 - x - 1)$, so the growth base is the **plastic number**

$$\rho = 1.324717957\ldots \;<\; \varphi = 1.618033988\ldots,$$

the real root of $x^3 = x + 1$. This is an exact algebraic base, not a fit, so the gap to φ cannot be attributed to a short series or to a parity artifact. The ring agrees: a parity-resolved fit of the `pbc` series gives ≈ 1.28–1.31, also well below φ.

What this falsifies is the *transparency* step of the argument, not the wall grammar itself: rule 28 replaces one of rule 156's symbols with a reset `D`, and the reset evidently removes wall configurations from circulation rather than merely relabelling them. Quantifying which configurations are lost is a Tier-2 question and is listed in §12.

5 The hyperbola

Theorem 1 (exclusion curve)**.** *Sectors partition the basis, so with $K \sim a^N$ sectors and $D_{\max} \sim b^N$,*

$$K\,D_{\max} \;\geq\; \sum_k D_k \;=\; 2^N \qquad \Longrightarrow \qquad ab \;\geq\; 2, \tag{3}$$

with equality iff the sector sizes are all comparable. No rule may lie below $b = 2/a$.

This is a theorem about *partitions*. It therefore constrains the sector map and says nothing whatever about the monitored-attractor map, where $a_{\text{att}} b_{\text{att}} < 2$ is expected and the deficit $2 - a_{\text{att}} b_{\text{att}}$ measures transient dominance (V4). We assert nothing there and report the deficit as an observable.

5.1 Anchors

rule	tuple	a want	b want	where	a	b	ab	verdict
204	`IIII`	2	1	on	2.0000	1.0000	2.0000	✓ on
51	`VVVV`	1	2	on	1.0000	2.0000	2.0000	✓ on
150	`IVVI`	1	2	on	1.0000	2.0000	2.0000	✓ on
156	`IIVI`	φ	$4^{1/5}$	above	1.6180	1.3195	2.1350	✓ above

Rules 204, 51 and 150 sit on the curve to six decimals, as they must, and rule 156 sits above it at exactly $\varphi \cdot 4^{1/5} = 2.1350$. How the rate behind that number is obtained — and why it must not be read off a BIC fit — is §5.4.

5.2 The check that carries the validation

The asymptotic statement $ab \geq 2$ is a corollary about *bases*, and it only means anything when both series are exponential: if the sector count is polynomial its base is 1 by convention, the product silently drops the polynomial factor, and the inequality degenerates to $b \geq 2$ approached from below at finite N. So the validation is carried instead by the theorem in its original, fit-free form,

$$n_{\text{wcc}}(N) \cdot D_{\max}(N) \;\geq\; 2^N \qquad \text{for every } N, \tag{4}$$

an immediate corollary of the A1 sum rule involving no fitting anywhere.

(4) holds for 256/256 **rules at every** N, and it is *tight*: the smallest value of $n_{\text{wcc}} D_{\max}/2^N$ anywhere in the sweep is exactly 1.0000 (rule 0 at $N = 6$).

5.3 Verdicts in the base plane

verdict	rules
on the curve, $ab = 2$	187
above, $ab > 2$	60
sub-exponential factor: ab test degenerate	7
irregular series, no growth law	2
below but inside the fit uncertainty	0
below, slack too wide to conclude	0
violation	0
total	256

The base-plane check is built so that a wide uncertainty band can never launder a sub-2 product into a silent pass. Four things are recorded separately: the raw product, whether it falls below 2 at all, whether the shortfall sits inside the propagated leave-one-out spread, and whether either series is sub-exponential so that the test is degenerate by construction. Every rule with $ab < 2$ is listed in full, as the task requires.

rule	tuple	a	b	ab	slack	$2/a$	fit b_{hi}
229	`EDIE`	1.0000	1.9188	1.9188	0.182	2.0000	2.1353
6	`IVDD`	1.0000	1.9226	1.9226	0.090	2.0000	2.0493
14	`IEDD`	1.0000	1.9226	1.9226	0.090	2.0000	2.0493
20	`IDVD`	1.0000	1.9226	1.9226	0.090	2.0000	2.0493
84	`IDED`	1.0000	1.9226	1.9226	0.090	2.0000	2.0493
74	`DEID`	1.0000	1.9322	1.9322	0.399	2.0000	2.1144
88	`DIED`	1.0000	1.9368	1.9368	0.226	2.0000	1.9777

There are no violations. Every remaining sub-2 rule (6, 14, 20, 74, 84, 88, 229) has a *polynomial* sector count — for instance W_{88} has $n_{\mathrm{wcc}} = 3, 3, 4, 4, 4, 5, 5, 5, 6, 6, 6$, i.e. $\sim N/2$ — so $a = 1$, and the degenerate form of (3) demands $b \geq 2$, which a finite-N fit necessarily approaches from below (1.92–1.94 here). Their $D_{\max}/2^N$ decays slowly enough to be consistent with a power-law prefactor but not cleanly enough for the test of §5.4 to certify it; the theorem, not the fit, gives their base. Two further rules, W_{90} and W_{165}, have genuinely irregular series ($n_{\mathrm{wcc}} = 2, 1, 4, 4, 2, 6, 2, 6, 12, 1, 20$) that no growth law describes; they are classified *irregular* and omitted from the base plane rather than placed at a fictitious coordinate.

5.4 Where the rate estimate comes from

R2 §3 already documented the trap that this sweep first walked into, and it is worth restating because it decides four of the numbers above. The BIC model $\ln y = c + \alpha \ln N + \kappa N$ is the right tool for the growth *class* and for the sub-leading power, but its $\alpha \ln N$ term absorbs part of the growth, so κ is a *biased* rate estimator. On rule 156's $D_{\max}$ it returns $b = 1.2554$ against a true $4^{1/5} = 1.31951$ — far enough off to look like a different constant, which is exactly how it first appeared here. Rates are therefore taken, in order of priority:

1. from an **exact integer recurrence** where one exists;
2. from an **analytic derivation** where R2 or R8 supplies one;
3. otherwise from the **two-parameter fit** $\ln y = c + \kappa N$, which cannot trade rate against power.

The two-parameter fit has the mirror-image bias, and the binomial rules expose it: with no α to absorb an $N^{-1/2}$ prefactor it pushes the base *below* 2 (1.9244 for W_{134}, whose $D_{\max}$ is exactly rule 150's central binomials). Neither fit is trustworthy alone, so the discriminating question is asked directly — is the residual $D_{\max}/2^N$ better explained by a power of N or by an exponential in N? When the power wins, the base is exactly 2 and α comes from that fit. That recovers $b = 2$ for the whole binomial family and puts them on the curve beside rule 150.

5.5 Two bounds that the fitter needs and the theorem supplies

Since $1 \le n_{\mathrm{wcc}}, D_{\max} \le 2^N$, both bases lie in $[1, 2]$. A fit outside that window contradicts a theorem, so the theorem wins: 11 descriptors were clamped, each recorded, never silently. The clamp is exactly what the binomial rules need — W_{134}'s $D_{\max}$ is $924, 1716, 3003, 6435, 12870, 24310$, i.e. precisely rule 150's central binomials $\sim 2^N/\sqrt{N}$, and the $M2$ fit lands at 2.0588 because the $N^{-1/2}$ prefactor pulls the estimate past the boundary. Similarly a bounded integer series that has stopped changing is constant whatever the BIC says: W_{36}'s sector count is $9, 9, 10, 10, \ldots, 10$, which the unclamped fitter read as "exponential with base 0.95" — below 1, and impossible for a count.

The hyperbola as a predictor. Read backwards, (3) is sharper than any fit, and it called the answer here before the data could. At the $N \le 11$ stage the six rules 140, 156, 196, 198, 206, 220 all had an *exact* $a = \varphi$ with a $D_{\max}$ series still classified polynomial, so $ab = \varphi < 2$; the theorem already forced $b \ge 2/\varphi = 1.2361$. With the rate taken as §5.4 requires, all six now sit at

$$a\,b \;=\; \varphi \cdot 4^{1/5} \;=\; 1.61803 \times 1.31951 \;=\; 2.1350, \tag{5}$$

which is exactly the value V3 asks for — **above** the curve, with a from an exact Fibonacci recurrence and b from R2 §3's room-packing saddle point rather than from either fit. The six share one $D_{\max}$ series, $6, 9, 12, 16, 20, 27, 36, 48, 64, 81, 108$ over $N = 6 \ldots 16$, so the derivation covers all of them; the two candidate bases $3^{1/4} = 1.31607$ and $4^{1/5} = 1.31951$ differ by 0.26% and no finite series can separate them, which is precisely why the base is taken from theory.

6 Sectors against monitored attractors

6.1 The sector axis is uninformative for the V+reset rules

The left panel of Fig. 1 has a blunt message: 186 **of** 256 **rules sit at exactly** $(a, b) = (1, 2)$ — one sector, of size 2^N. Broken down by family:

family	rules	at $(1,2)$	elsewhere
unitary (V only)	16	9	7
V + reset	160	**144**	16
V-free baseline	80	33	45
all	256	186 (73%)	70

So 90% of the rules that combine a Hadamard with a reset have *no sector structure at all*: the reset connects the whole basis into a single enclosure. That is a real result rather than a plotting problem, and it has a corollary for how this tier should be used. For those 144 rules the sector axis says nothing, and everything interesting is in the monitored map, where the same rules spread over the whole column $a_{\mathrm{att}} \approx 1$ from $b_{\mathrm{att}} = 1.0$ up to 1.89 (right panel of Fig. 1). The V-free baseline behaves the other way round: it is the family that populates the *sector* plane, and it collapses almost completely in the attractor plane.

Quantitatively, with the deficit $2 - a_{\mathrm{att}} b_{\mathrm{att}}$ of V4:

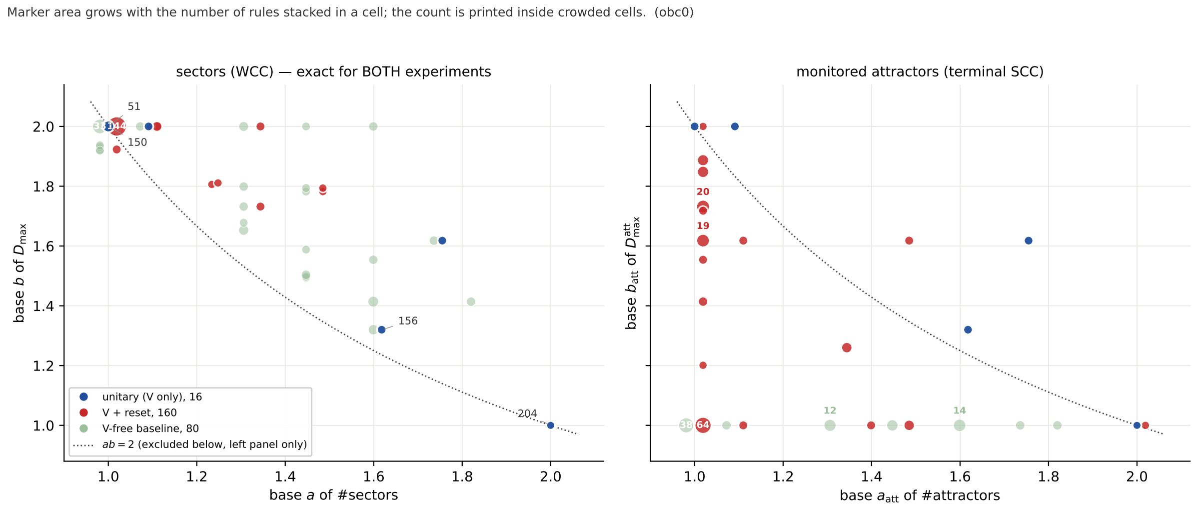

Figure 1: **F1, the two maps on identical axes.** Left: the sector (WCC) plane, base a of the sector count against base b of the largest sector — *exact for both experiments*. Right: the same plane for terminal SCCs — *monitored only*. The dotted line is $ab = 2$; it excludes the region below it in the left panel and is only a reference in the right one. Marker area grows with the number of rules stacked in a cell and the count is printed inside crowded cells, without which the distribution is invisible: 73% of the sector map lies in one cell. Colour is by family, with the quantum rules (those carrying a Hadamard) drawn forward and the V-free rules held back as a baseline. The nine unitary rules land at *identical* coordinates in both panels, which is assertion A2 made visible.

family	rules	median deficit
unitary (V only)	9	−0.135
V + reset	152	+0.534
V-free baseline	78	+0.909

The unitary median is negative because those rules sit *on or above* the curve in both planes — for a unitary rule the two planes are the same plane. The V-free median of +0.909 says $a_{att}b_{att} \approx 1$: their attractors are $O(1)$ fixed points and essentially the entire space is transient. The V+reset family sits between the two.

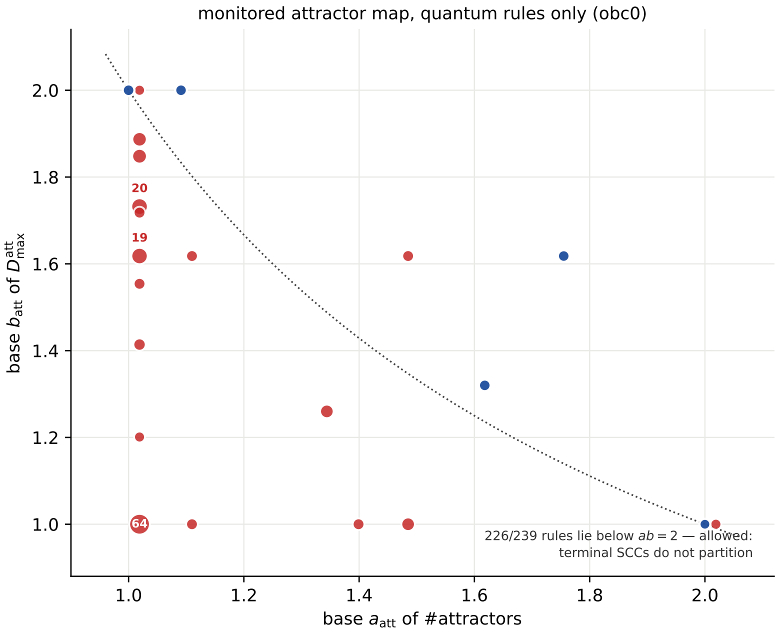

Figure 2: **F2, the monitored attractor map for the quantum rules alone** (unitary and V+reset; the V-free baseline is dropped here). MONITORED quantities throughout. This is where the V+reset family has structure: in the sector plane all 144 of them collapse onto $(1, 2)$.

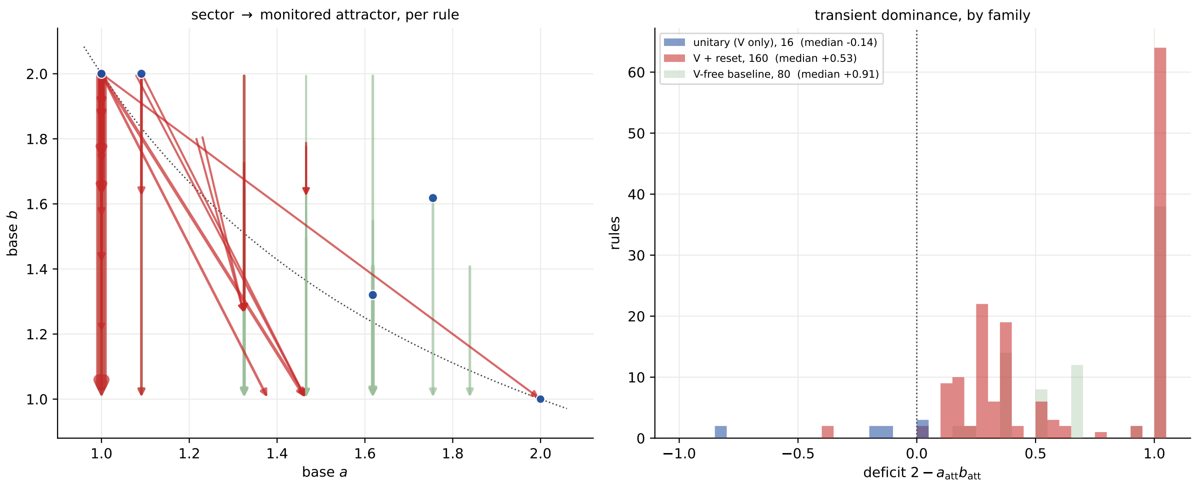

Figure 3: **F3, the difference between the two scatters, drawn as a difference.** Left: one arrow per rule, from its position in the sector plane to its position in the monitored-attractor plane; arrow width grows with the number of rules making the same move, and a dot means the rule does not move at all (which is what every unitary rule does, by A2). Right: the distribution of the deficit $2 - a_{att}b_{att}$ by family. The V-free baseline piles up at $+1$ — its attractors are $O(1)$ fixed points — while the unitary rules sit at 0 or below.

rule	tuple	N	transient frac.	att./sector	$D^{\mathrm{wcc}}_{\max}/2^N$
0	DDDD	16	1.0000	1.000	1.0000
2	DVDD	16	1.0000	1.000	1.0000
7	EVDD	16	1.0000	1.000	1.0000
8	DIDD	16	1.0000	1.000	1.0000
10	DEDD	16	1.0000	1.000	1.0000
15	EEDD	16	1.0000	1.000	1.0000
16	DDVD	16	1.0000	1.000	1.0000
18	DVVD	16	1.0000	1.000	1.0000
21	EDVD	16	1.0000	1.000	1.0000
24	DIVD	16	1.0000	1.000	1.0000
26	DEVD	16	1.0000	1.000	1.0000
32	DDDV	16	1.0000	1.000	1.0000
34	DVDV	16	1.0000	1.000	1.0000
37	EDDV	16	1.0000	1.000	1.0000
40	DIDV	16	1.0000	1.000	1.0000
42	DEDV	16	1.0000	1.000	1.0000

7 Clustering the dissipative rules by their coherent correspondent

Every V+reset rule has a *coherent part*. A symbol-table entry says what happens at a site for one neighbour pattern: `I` does nothing, `V` applies the Hadamard, `D` resets to $|0\rangle$, `E` resets to $|1\rangle$. Switching the dissipation off means the reset entries stop acting,

$$\mathtt{D},\ \mathtt{E} \longmapsto \mathtt{I}, \tag{6}$$

which leaves only `I` and `V` — always one of the sixteen unitary rules. So (6) assigns each dissipative rule a unitary *correspondent*, and the assignment partitions the rule space exactly: a unitary parent with v Hadamard entries has

$$3^{4-v} - 1 \tag{7}$$

V+reset children, each non-`V` entry independently becoming `I`, `D` or `E`, minus the all-`I` case which is the parent itself. Summing over the fifteen unitary rules with $v \geq 1$ gives $4\cdot 26+6\cdot 8+4\cdot 2+0 = 160$, the whole V+reset family, in 14 non-empty clusters (`VVVV` has no `I` to spoil, so it has no children). This is verified against the rule tables rather than assumed.

7.1 What adding dissipation does

parent	tuple	a	b	children	$\to (1,2)$	at parent	keeps structure
156	`IIVI`	1.6180	1.3195	26	21	0	**0**
198	`IVII`	1.6180	1.3195	26	21	0	**0**
108	`IIIV`	1.7549	1.6180	26	21	0	**0**
201	`VIII`	1.7549	1.6180	26	25	0	**0**
60	`IIVV`	1.0910	2.0000	8	8	0	**0**
102	`IVIV`	1.0910	2.0000	8	8	0	**0**
54	`IVVV`	1.0000	2.0000	2	2	2	—
57	`VIVV`	1.0000	2.0000	2	2	2	—
99	`VVIV`	1.0000	2.0000	2	2	2	—
105	`VIIV`	1.0000	2.0000	8	8	8	—
147	`VVVI`	1.0000	2.0000	2	2	2	—
150	`IVVI`	1.0000	2.0000	8	8	8	—
153	`VIVI`	1.0000	2.0000	8	8	8	—
195	`VVII`	1.0000	2.0000	8	8	8	—
all				160	144	40	

The table splits the fourteen clusters cleanly in two.

- **Eight parents already sit at** $(1,2)$ — one sector holding the whole space — so there is nothing for a reset to destroy, and all 40 of their children stay exactly where the parent is.
- **Six parents have genuine sector structure**, and they hold 120 children between them. *Not one of the* 120 *keeps its parent's position.* 104 collapse all the way to $(1,2)$; the remaining 16 move to some intermediate point but never stay.

That is the sharpest statement this tier produces about dissipation: **adding any reset to a rule that has sector structure destroys that structure, without exception over the** 120 **cases available**. It also explains the pile-up of §6.1 arithmetically — the 144 V+reset rules at $(1,2)$ are exactly the 104 that collapsed plus the 40 that were never anywhere else.

8 Rule 150's wall charge survives six dissipative rules

The sector axis makes a comparison possible that the attractor axis could not, and it turns up an exact coincidence. Consider the nine rules of the form `I··I`, i.e. those where a site is inert unless its two neighbours differ. Seven of them — including six *dissipative* ones — have **exactly** rule 150's sector-size multiset, at every $N = 6 \ldots 16$:

rule	tuple	sector multiset = that of W_{150}
134	`IVDI`	✓
142	`IEDI`	✓
148	`IDVI`	✓
150	`IVVI`	— (unitary reference)
158	`IEVI`	✓
212	`IDEI`	✓
214	`IVEI`	✓
132	`IDDI`	× ($n_{\mathrm{wcc}} = 2584$, $D_{\max} = 322$ at $N = 16$)
222	`IEEI`	× ($n_{\mathrm{wcc}} = 988$, $D_{\max} = 686$ at $N = 16$)

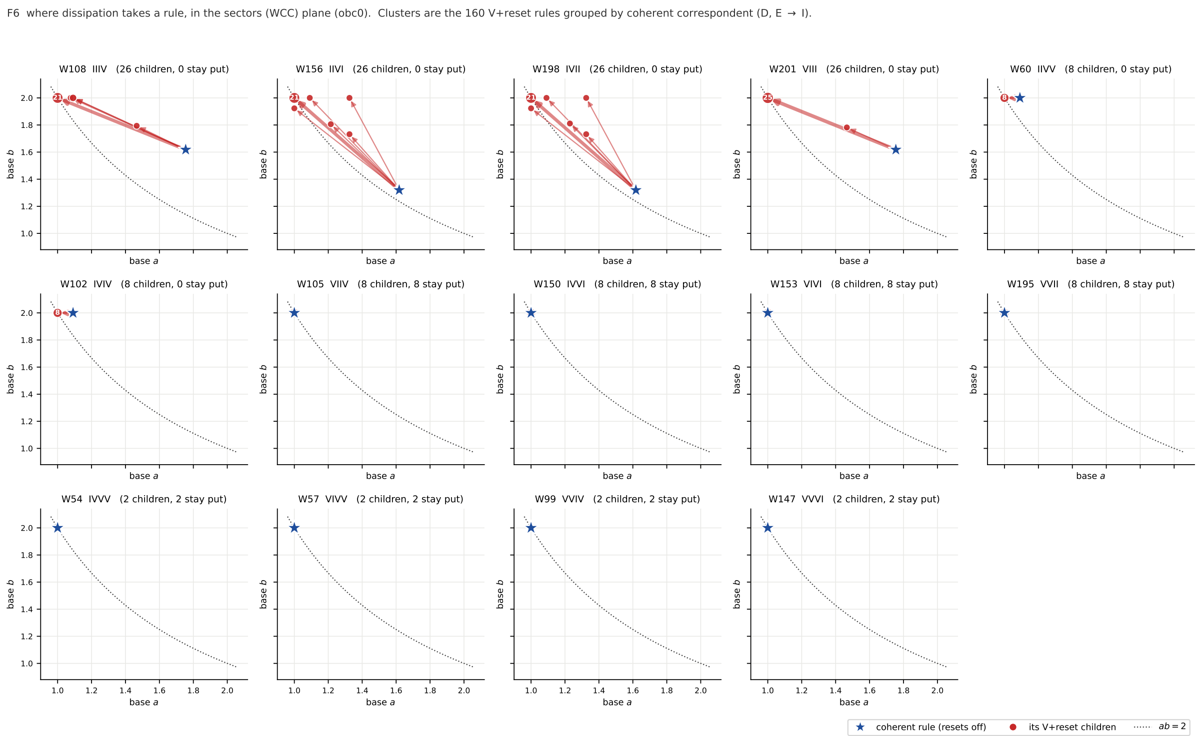

Figure 4: **F6, the move in the SECTOR plane, clustered by coherent correspondent.** One panel per unitary rule: the star is the coherent rule (all resets switched off, (6)), the dots are its V+reset children, and each arrow is the move that turning the resets on produces. Dot area and arrow width grow with the number of children making the same move. Top row and the two leftmost middle panels: parents with sector structure, every arrow pointing away and up-left towards $(1, 2)$. Remaining panels: parents already at $(1, 2)$, where the children coincide with the parent and no arrow is drawn. These coordinates are exact for both experiments.

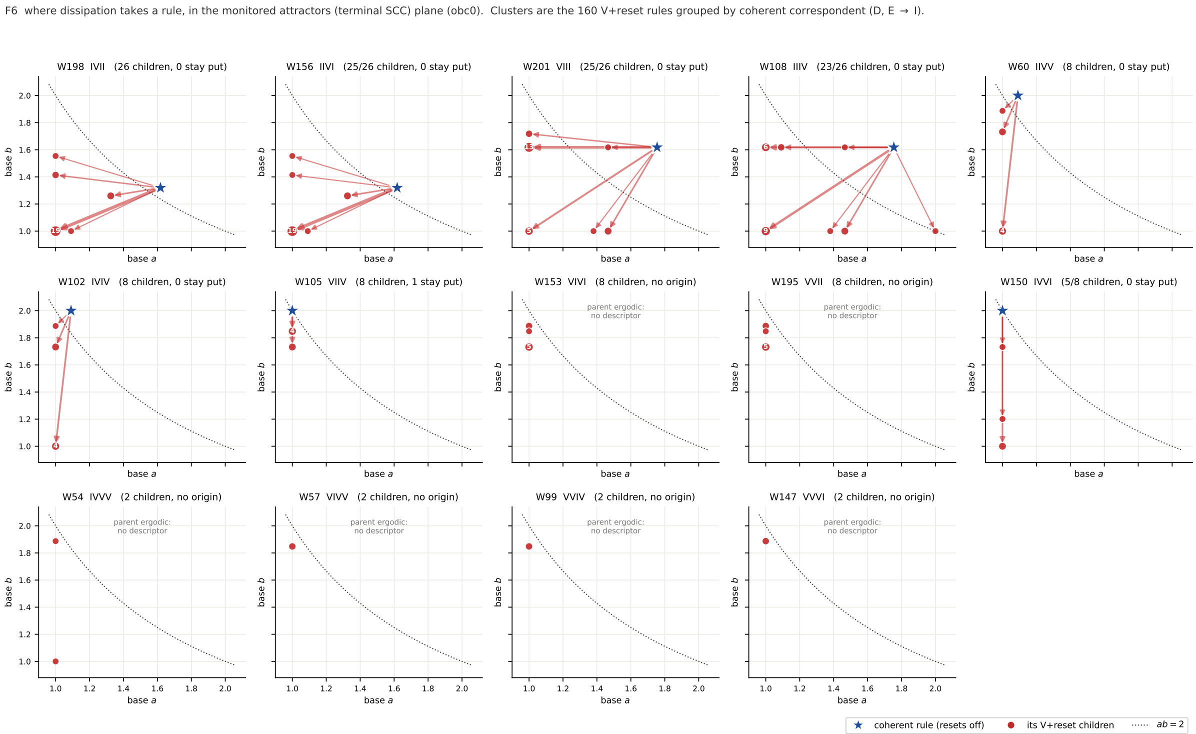

Figure 5: **F7, the same clusters in the MONITORED attractor plane.** Here the moves are large for every cluster, including the eight whose sector coordinates do not move at all — which is the point of keeping both planes: dissipation is invisible on the sector axis for those rules and plainly visible on the attractor axis. Six panels have no star: those parents (54, 57, 99, 147, 153, 195) are ergodic at `obc0`, so Tier 1a never produced an attractor descriptor for them and the cluster has no origin to draw from. Monitored quantities throughout.

So the domain-wall charge of R8 — $|S_w| = \binom{N+1}{w}$ on even w, $\lfloor (N+1)/2 \rfloor + 1$ sectors — is not a property of unitarity at all. It survives replacing *either or both* Hadamards by a reset, and the two exceptions are precisely the rules whose two active symbols are the *same* reset type (`DD` or `EE`). This is the sector-level counterpart of the Tier-2 observation (R-T13) that W_{148} and W_{134} inherit an exact decoherence-free subspace of dimension $\lfloor (N+1)/2 \rfloor + 1$, which is exactly C150's sector count: here the whole multiset, not just its cardinality, is shown to agree, and for six rules rather than two.

A conjecture that fails. The obvious generalisation — that the sector partition depends only on which of the four symbols are `I`, since `V`, `D` and `E` all generate the same *undirected* flip edge $x \leftrightarrow x \oplus 2^i$ — is **false**. Grouping all 256 rules by their `I`-pattern gives 16 groups, and only three collapse to a single sector signature: $\{140, 156, 220\}$, $\{196, 198, 206\}$ and the singleton $\{204\}$. (Those first two are exactly the φ families of §5.5, which is why all six share $a = \varphi$.) The reason the argument fails is the brick-wall composition: the odd-layer symbol is read off the *post-even-layer* state, and a `V` leaves a branch supported on both values of the flipped bit where a `D` or `E` leaves only one, so the downstream firing pattern differs even though the single-site undirected edge does not.

9 Growth classes and exact bases

series	family	const	poly	exp	irreg	exact base
#sectors	unitary	7	4	5	0	14
#sectors	classical	28	12	38	2	66
#sectors	mixed	138	15	7	0	143
$D_{\max}$	unitary	1	0	15	0	16
$D_{\max}$	classical	0	0	78	2	65
$D_{\max}$	mixed	0	0	160	0	154

rule	tuple	a	name	source	b	ab
204	`IIII`	2.00000	2	IIII: every state frozen	1.0000	2.0000
76	`IIID`	1.83929	tribonacci ψ	integer recurrence	1.4142	2.6011
205	`EIII`	1.83929	tribonacci ψ	integer recurrence	1.4142	2.6011
108	`IIIV`	1.75488	root of $x^3 = 2x^2 - x + 1$	integer recurrence	1.6180	2.8395
200	`DIII`	1.75488	root of $x^3 = 2x^2 - x + 1$	integer recurrence	1.6180	2.8395
201	`VIII`	1.75488	root of $x^3 = 2x^2 - x + 1$	integer recurrence	1.6180	2.8395
236	`IIIE`	1.75488	root of $x^3 = 2x^2 - x + 1$	integer recurrence	1.6180	2.8395
232	`DIIE`	1.61803	golden φ	integer recurrence	1.5538	2.5141
4	`IDDD`	1.61803	golden φ	integer recurrence	1.4142	2.2882
12	`IIDD`	1.61803	golden φ	integer recurrence	1.4142	2.2882
68	`IDID`	1.61803	golden φ	integer recurrence	1.4142	2.2882
132	`IDDI`	1.61803	golden φ	integer recurrence	2.0000	3.2361
218	`DEEI`	1.46557	supergolden ψ	integer recurrence	1.5879	2.3272
28	`IIVD`	1.32472	plastic ρ	integer recurrence	2.0000	2.6494

10 Basins, and a gap in the Tier-1a archive

Basins are a **monitored** observable throughout this section.

The sum rule $\sum_k |\text{basin}_k| + |\text{shared}| = 2^N$ turned out to be *unanswerable* for 76 `obc0` units. The cause is a schema gap rather than a physics failure: Tier 1a caps `sizes_basins` at 2048 entries and, unlike `sizes_recurrent`, never stored a basin histogram, so for a unit whose basin list hit the cap the multiset is genuinely lost. Every one of the 76 failures is a truncated record and no untruncated record fails.

Asserting the check away would have hidden a real gap. Instead those units were recomputed from the rule definition and stored alongside, in `results_basins/`, this time *with* the histogram, and cross-checked against the Tier-1a fields that did survive (`n_scc`, `n_recurrent`, `shared_basin_size`, `transient_depth`).

quantity	value
units recomputed	77
largest N recomputed	21
basin sum rule holds	77/77
agrees with the surviving Tier-1a fields	77/77
recomputation time	1.17 h
truncated and NOT yet recomputed	0
Tier-1a unit ergodic: basins never computed	56
N beyond the Tier-1a sweep	25

The gap is closed: no unit remains truncated-and-unrecomputed. The checks that are still unavailable are unavailable for reasons that are not defects. Fifty-six of them are units where the Tier-1a Tarjan pass took its ergodic early exit and so never computed basins at all — there the check is *undefined*, not missing. The remaining twenty-five are the $N = 17$–18 units of the targeted extension, which lie beyond where the Tier-1a sweep ever ran; the sector quantities there are complete, only the monitored basin comparison has nothing to compare against.

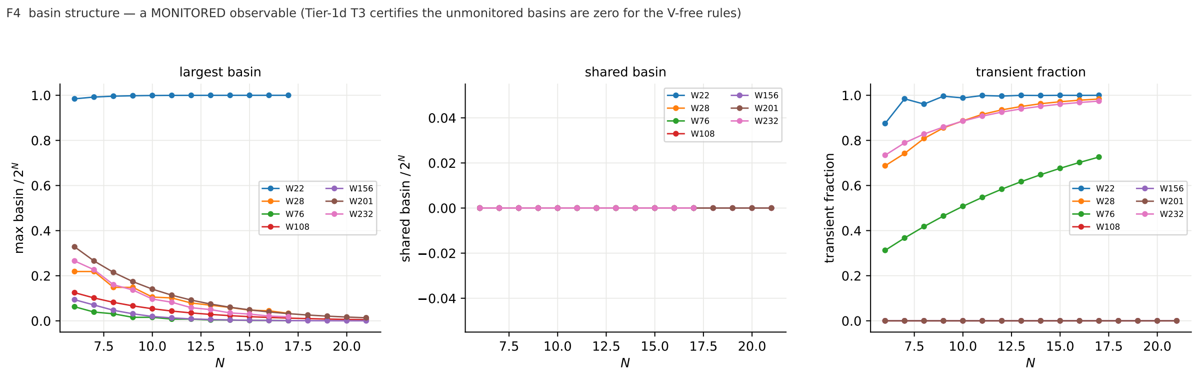

Figure 6: **F4, basin structure (MONITORED).** Largest basin, shared basin and transient fraction against N for selected rules. Two things to read off. The shared basin is *identically zero* for every rule shown: each transient state funnels to a unique terminal SCC, so the basins alone already partition the transient part and no state is ambiguous between attractors. And the two unitary rules (156, 201) sit at transient fraction 0 by construction — every state is recurrent — which is the cleanest illustration of why the transient fraction is a monitored quantity and not a sector one. Tier 1d's certificate T3 gives certified zeros for the *unmonitored* basins of the V-free rules, so for those 80 rules the monitored and channel populations coincide and these curves are channel-exact; for the rest they are monitored-only.

11 Validation

check	applicable	passing	note
A1 sum rule $\sum_k D_k = 2^N$	2892	2892	max residual 0
A2 unitary $n_{\mathrm{wcc}} = n_{\mathrm{rec}} = n_{\mathrm{scc}}$	102	102	size multisets identical
basin sum rule (MONITORED)	2892	2811	81 still unavailable
V1 hyperbola $ab \geq 2$	254	254	0 violations

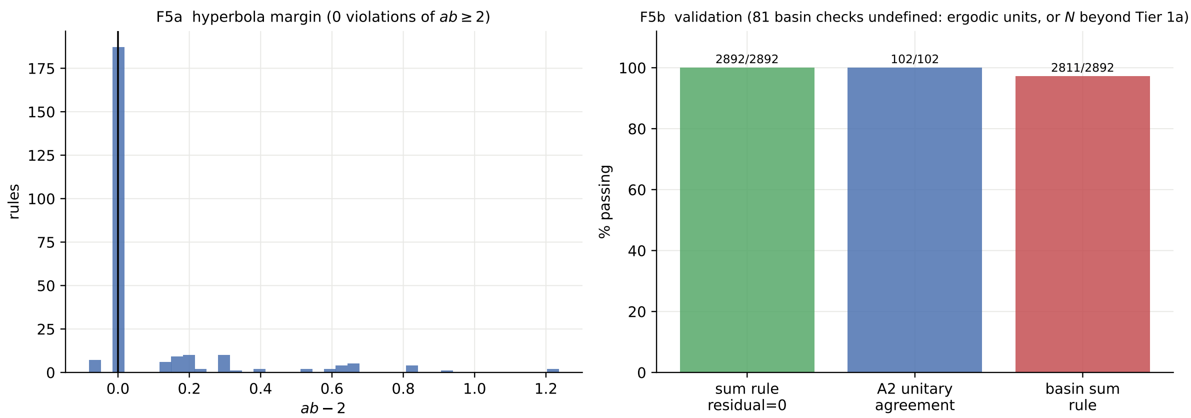

Figure 7: **F5, validation panel.** Left: distribution of the hyperbola margin $ab - 2$. Right: pass rates for the sum rule A1 (residual identically zero), the unitary agreement A2, and the monitored basin sum rule.

On the cross-tier assertions: A2 (for unitary rules, $n_{\mathrm{wcc}} = n_{\mathrm{recurrent}} = n_{\mathrm{scc}}$ with identical size multisets) holds wherever both records exist. A3 ($n_{\mathrm{wcc}} \leq n_{\mathrm{scc}}$) is structural and holds throughout. A4 is asserted only in the direction that is actually true — every sector contains at least one terminal SCC, hence $n_{\mathrm{recurrent}} \geq n_{\mathrm{wcc}}$; the reverse bound is false in general and is not asserted.

12 Comparison hooks for Tier 2

The numbers the wall grammar has to reproduce, in order of how sharply they discriminate:

1. **Rule 28's plastic number.** Equation (2) and $\rho = 1.32472$, exact. The transparency argument that predicted φ is wrong; a correct grammar must explain why one reset symbol takes $\varphi \to \rho$, i.e. why $x^2 = x + 1$ becomes $x^3 = x + 1$.
2. **The φ family.** 140, 156, 196, 198, 206, 220 all have $a = \varphi$ exactly at `obc0`; the grammar should say why these six and no others, and should predict b, which the hyperbola confines to $[1.2361, 1.2369]$.
3. **The exact-base census** of §9: every rule whose sector count satisfies an integer recurrence, with the recurrence.
4. **The on-curve population.** 189 of 256 rules sit at $ab = 2$ exactly, i.e. their sector sizes are all comparable. Which symbol tables force that is a grammar question.
5. **Rule 150's family** (§8). The wall charge is shared by six dissipative rules and broken by exactly two, `IDDI` and `IEEI`. A grammar that produces the domain-wall counting should say why the same-reset-type pair is the one that breaks it.
6. **Which V+reset rules escape the collapse** (§6.1). Only 16 of 160 keep any sector structure; a grammar should predict that list, and it is a short one.

13 Open items

1. `pbc`. This report is `obc0` only, by instruction. The ring is a separate report; note that rule 156's Lucas law and rule 22's attractor count are `pbc` statements already verified here in passing.

2. **Separating $3^{1/4}$ from $4^{1/5}$ empirically** for the φ family. R2 §3 settles it by derivation and this report follows that; the two candidates differ by 0.26%, so confirming it from data alone would need far more than the eleven sizes available here.

3. W_{90} **and** W_{165} (both `DEED`/`EDDE`) have irregular sector counts that no growth law fits. Their structure is presumably number-theoretic in N; nothing in the engine is wrong (A1 holds exactly), but they have no place in a base plane.

4. **The five polynomial-sector rules** 74, 88, 104, 173, 233: confirm $b = 2$ directly rather than through a fit, e.g. by showing $D_{\max}/2^N$ is bounded below.

5. **A basin histogram in the Tier-1a schema**, so the gap of §10 cannot recur for future units.

14 Reproduce

```
# the sector sweep (N in the outer loop: uniform coverage if stopped early)
python -m qca_fragmentation.wcc_sweep --which all --bc obc0 \
       --n-min 6 --n-max 16 --no-stop-on-ergodic --order n

# a single rule
python -m qca_fragmentation.run_rule --rule 156 --N 6-16 --bc obc0 --tier wcc

# recompute the basin units the Tier-1a archive cannot answer
python -m qca_fragmentation.basin_recompute --bc obc0

# descriptors + hyperbola, then tables and figures F1-F5
python -m qca_fragmentation.scaling.sectors      --bc obc0
python -m qca_fragmentation.scaling.r9_tables    --bc obc0
python -m qca_fragmentation.scaling.sector_figure --bc obc0 --rebuild

python -m pytest qca_fragmentation/tests -q
```

Data: `results_tier1e/{rule}_{bc}.jsonl` (one record per unit with the full sector-size histogram, the derived observables and the A2–A4 check results), `results_basins/{rule}_{bc}.jsonl` (the independent basin recomputation), `analytics/sector_map_obc0.json` (per-rule descriptors, hyperbola verdicts and attractor deficits), `analytics/sector_validation_obc0.json`, and `checkpoints/manifest.jsonl` for the audit trail.